

Statistical Methods in Chemistry

Block 3

Calibration Curves and Linear Regression

Student handout

Name: _____

Class: _____

Date: _____

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1 Why calibration curves matter

Many chemical measurements do not directly give concentration. A colorimeter gives absorbance, a conductivity probe gives conductivity, and a sensor gives a signal. To convert a signal into a concentration, chemists compare the unknown sample with standards of known concentration.

Core idea

A **calibration curve** is a graph made using known standards. Once the relationship between concentration and signal is known, the signal from an unknown sample can be converted into an estimated concentration.

The usual workflow is:

- 1) Prepare a blank and several standards of known concentration.
- 2) Measure the instrument signal for each standard.
- 3) Plot signal against concentration.
- 4) Fit a suitable model, usually a straight line over a limited range.
- 5) Use the equation of the line to calculate the concentration of an unknown.

Term	Meaning in a calibration experiment
Standard	A solution with known concentration.
Blank	A solution containing everything except the analyte; used to check background signal.
Independent variable	The variable chosen by the experimenter, usually concentration of the standards.
Dependent variable	The measured signal, such as absorbance.
Calibration equation	The fitted equation that converts between concentration and signal.
Unknown	A sample whose concentration is estimated using the calibration curve.

The graph is not decoration

A calibration curve is part of the measurement method. If the calibration is weak, the concentration calculated for the unknown is also weak, even if the arithmetic is correct.

Quick exercise: Variables in calibration

For each experiment, identify the independent variable and dependent variable.

- a) Copper(II) sulfate standards are measured using a colorimeter.
- b) Nitrate standards are measured using an ion-selective electrode.
- c) Standard glucose solutions are measured using a glucose meter.

2 Constructing a calibration curve

A calibration curve should usually be drawn as a **scatter plot**, not as a line graph connecting the dots. The standards are experimental data points. The best-fit line represents the estimated relationship.

Worked example: Copper sulfate colorimetry data

A student prepares copper sulfate standards and measures their absorbance.

Concentration, C / mg L^{-1}	Absorbance, A
0	0.012
20	0.133
40	0.247
60	0.374
80	0.489
100	0.613

Linear regression gives approximately

$$A = 0.00600C + 0.0113.$$

This means that absorbance increases by about 0.00600 for each 1 mg L^{-1} increase in concentration. The intercept, 0.0113, estimates the background absorbance when the concentration is zero.

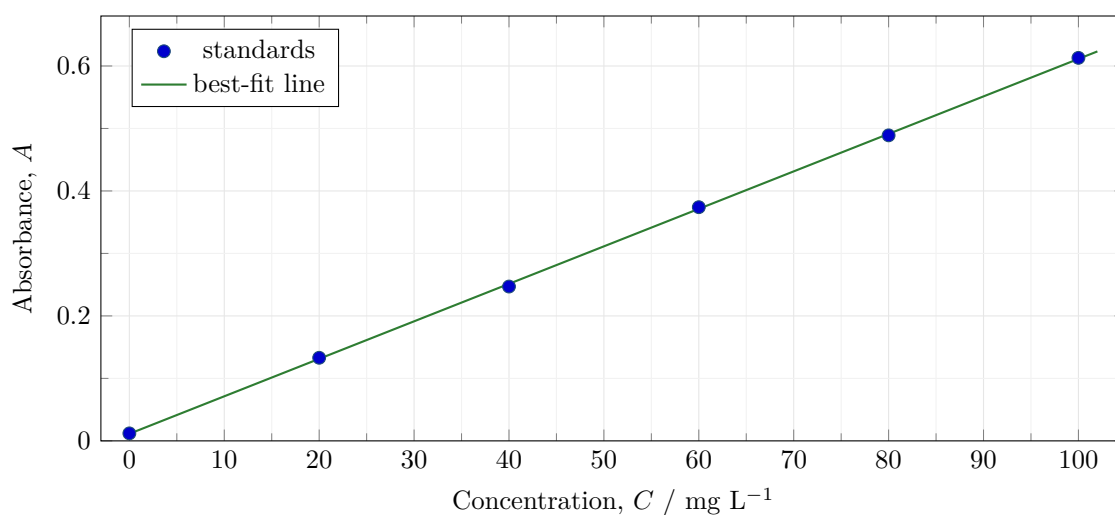


Figure 1. A calibration curve should show individual standard points and the fitted line.

Quick exercise: Reading the calibration equation

Use the calibration equation

$$A = 0.00600C + 0.0113.$$

- Predict the absorbance of a 75 mg L^{-1} standard.
- Estimate the absorbance of the blank.
- Calculate the concentration of an unknown with absorbance $A = 0.493$.

3 Linear regression: slope, intercept, and prediction

For a straight-line calibration, the fitted equation is written as

$$\hat{y} = mx + b.$$

Here x is the known concentration, y is the measured signal, and $\hat{y}$ is the signal predicted by the model. The slope m describes sensitivity. The intercept b describes the predicted signal when $x = 0$.

Regression line

For data points (x_i, y_i) , linear regression chooses the line that minimizes the sum of squared residuals:

$$\sum (y_i - \hat{y}_i)^2.$$

The fitted parameters are

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad b = \bar{y} - m\bar{x}.$$

In normal laboratory work, these values are usually calculated using a spreadsheet or statistical software, not by hand.

Worked example: Using a calibration equation for an unknown

Suppose the fitted equation is

$$A = 0.00600C + 0.0113.$$

An unknown sample gives $A = 0.342$. Rearrange the equation:

$$C = \frac{A - b}{m} = \frac{0.342 - 0.0113}{0.00600} = 55.1 \text{ mg L}^{-1}.$$

The estimated concentration of the measured solution is therefore about 55.1 mg L^{-1} .

Quick exercise: Two-point approximation

A rough calibration line passes through $(20, 0.128)$ and $(80, 0.486)$, where x is concentration in mg L^{-1} and y is absorbance.

- Calculate the slope.
- Calculate the intercept.
- Estimate the concentration of an unknown with absorbance $A = 0.300$.

4 Residuals: checking the fit

A line may look convincing, but a chemist should still check how far each standard lies from the fitted line. This difference is called a **residual**.

Residual

For each point,

$$\text{residual} = y_i - \hat{y}_i.$$

A positive residual means the measured signal is above the fitted line. A negative residual means it is below the fitted line.

For the copper sulfate data, the residuals are small and do not show a strong curved pattern. That supports the use of a straight-line calibration over this concentration range.

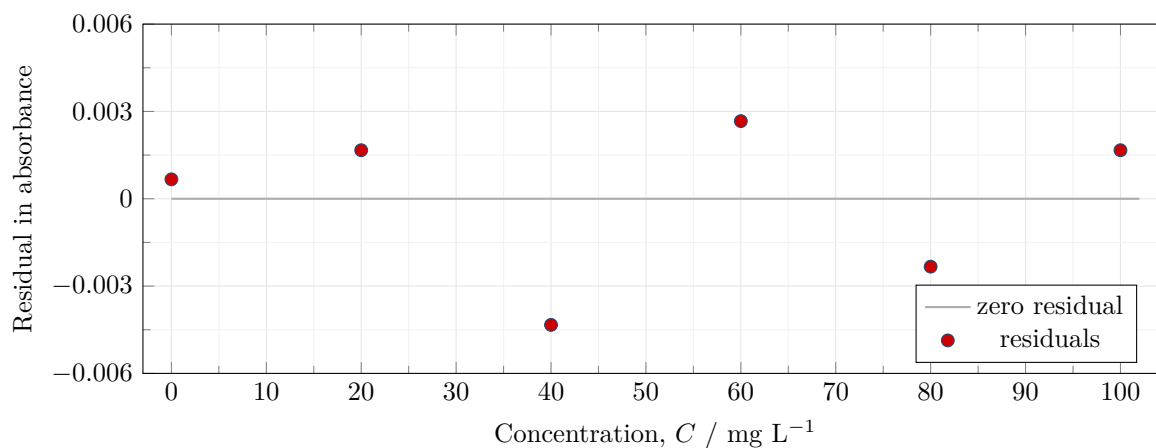


Figure 2. A residual plot should scatter around zero. A systematic pattern suggests that a straight line may not be appropriate.

Quick exercise: Residual calculation

The fitted equation is $A = 0.00600C + 0.0113$. A 40 mg L^{-1} standard has measured absorbance $A = 0.247$.

- Calculate the predicted absorbance $\hat{A}$.
- Calculate the residual $A - \hat{A}$.
- Is the point above or below the line?

5 The coefficient of determination, R^2

The value R^2 describes how much of the variation in the measured signal is explained by the fitted model. For a calibration curve, a value close to 1 often indicates a strong linear relationship.

R^2

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}.$$

An R^2 value of 1 means the fitted line passes exactly through all the data points. An R^2 value near 0 means the line explains little of the variation in the data.

For the copper sulfate data in Figure 1,

$$R^2 \approx 0.99985.$$

This is strong evidence that the standards are highly linear in this range. However, R^2 alone is not enough.

R^2 is not a complete validation

A high R^2 does not prove that the calibration is valid. You still need to check the blank, residuals, concentration range, possible outliers, and whether the unknown lies inside the standard range.

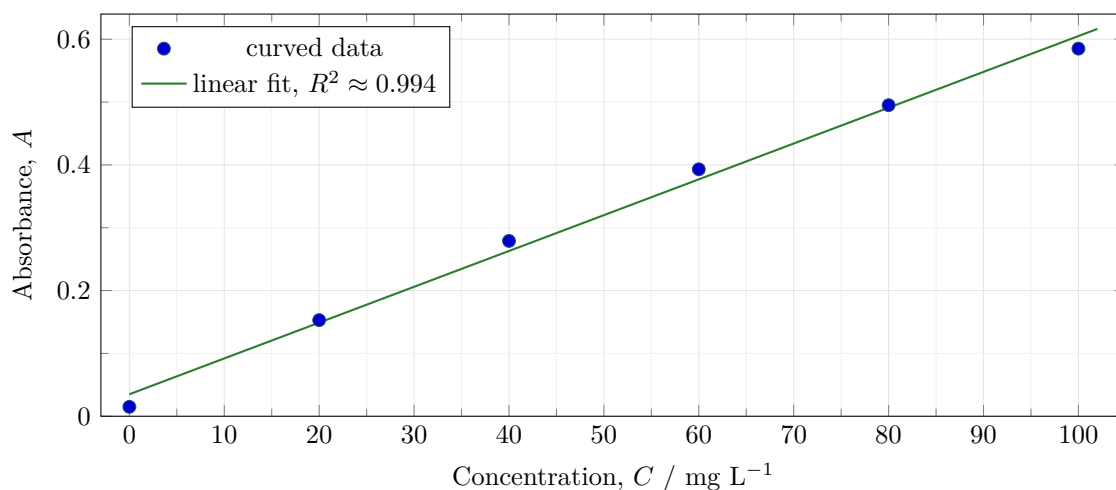


Figure 3. A high R^2 can still hide curvature. Residuals and chemical reasoning must be checked.

Quick exercise: Interpreting R^2

A calibration curve has $R^2 = 0.997$. Decide whether each conclusion is justified.

- The standards are probably close to a straight-line trend.
- The unknown concentration is definitely correct.
- The method has no systematic error.
- The residual plot should still be checked.

6 Interpolation and extrapolation

A calibration curve is most reliable inside the range covered by the standards. Estimating an unknown inside the standard range is called **interpolation**. Estimating outside the standard range is called **extrapolation**.

Interpolation versus extrapolation

Interpolation uses the calibration model inside the measured standard range. This is usually acceptable if the calibration is good.

Extrapolation uses the model outside the measured standard range. This is risky because the relationship may stop being linear beyond the standards.

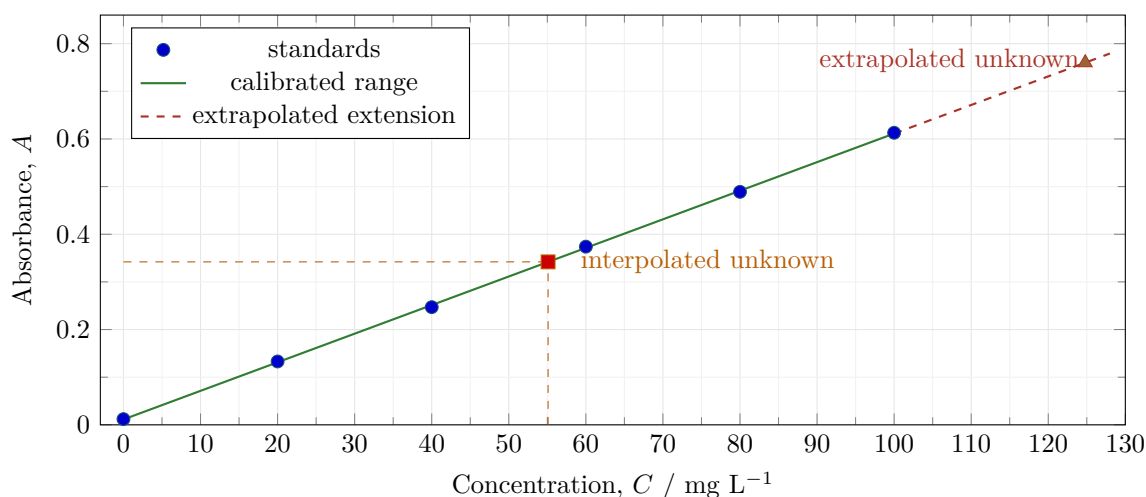


Figure 4. The orange unknown lies inside the standard range. The red unknown lies outside the range and should normally be diluted and remeasured.

Quick exercise: Interpolation or extrapolation?

The standards in Figure 4 cover approximately $A = 0.011$ to $A = 0.613$. Classify each unknown signal as interpolation, extrapolation, or questionable.

- $A = 0.302$
- $A = 0.005$
- $A = 0.640$
- $A = 0.488$

7 Spreadsheet workflow

Manual regression calculations are useful for understanding, but most real calibration work is done using a spreadsheet or statistical software. The main danger is not the calculation itself, but entering the ranges incorrectly or accepting a graph without checking it.

Task	Spreadsheet formula or action
Make the graph	Insert scatter plot with concentration on the x-axis and signal on the y-axis.
Slope	<code>=SLOPE(absorbance_range, concentration_range)</code>
Intercept	<code>=INTERCEPT(absorbance_range, concentration_range)</code>
R^2	<code>=RSQ(absorbance_range, concentration_range)</code>
Unknown C	<code>=(unknown_signal - intercept)/slope</code>
Residual	<code>=measured_signal - predicted_signal</code>

Common spreadsheet mistakes

Do not swap the x and y ranges, use a line chart instead of a scatter plot, connect standards point-to-point, over-round the trendline equation, or calculate an unknown outside the calibrated range without dilution.

Quick exercise: Spreadsheet formula check

In a spreadsheet, concentrations are in cells A2:A7, absorbances are in B2:B7, the slope is stored in E2, the intercept is stored in E3, and an unknown absorbance is in B10.

- Write a formula to calculate the slope.
- Write a formula to calculate the intercept.
- Write a formula to calculate the unknown concentration.

8 Mini-practical: determine an unknown concentration

A student prepares a calibration curve for a blue food dye solution. The unknown sample was diluted by transferring 10.00 mL of the original sample to a 50.00 mL volumetric flask and making it up to the mark.

Standard concentration / mg L^{-1}	Absorbance	Sample	Absorbance
0	0.003	Diluted unknown	0.268
10	0.079	Check sample	0.421
20	0.151		
30	0.229		
40	0.303		
50	0.378		

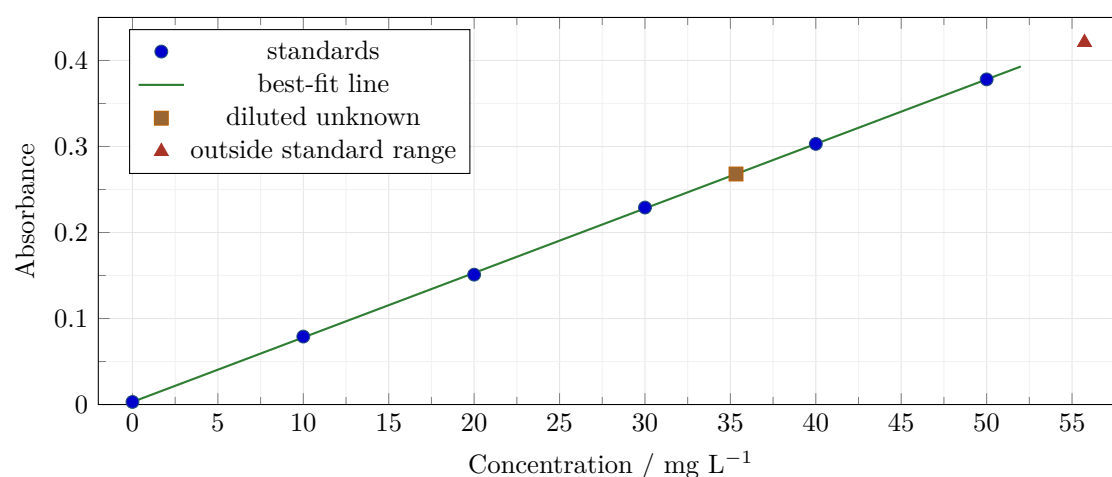


Figure 5. Mini-practical calibration curve. The diluted unknown is inside the range. The check sample is above the highest standard.

Quick exercise: Mini-practical questions

Use the data above.

- a) Record the calibration equation and R^2 from regression.
- b) Calculate the concentration of the diluted unknown.
- c) Calculate the concentration of the original unknown before dilution.
- d) Explain why the check sample should not simply be reported from this calibration.
- e) State one improvement to the calibration design.

9 Reporting calibration results

A good reported result should include both the calculated concentration and the evidence that the calibration was suitable.

Minimum reporting standard

For a calibration-based concentration, report:

- 1) the calibration equation;
- 2) R^2 ;
- 3) whether the unknown was inside the standard range;
- 4) the calculated concentration of the measured solution;
- 5) any dilution factor and the concentration of the original sample;
- 6) limitations, especially curvature, outliers, or extrapolation.

Worked example: Short reporting paragraph

The calibration equation was $A = 0.00750C + 0.00300$ with $R^2 = 0.99994$, where C is concentration in mg L^{-1} . The diluted unknown gave $A = 0.268$, which lies inside the standard range. The calculated concentration of the diluted solution was 35.3 mg L^{-1} . Since the original sample was diluted from 10.00 mL to 50.00 mL, the dilution factor was 5.00, giving an original concentration of approximately 177 mg L^{-1} . This result assumes that the calibration remains linear over the standard range and that the blank correction was valid.

Quick exercise: Write a conclusion

Write a concise conclusion for the copper sulfate unknown from Section 3. Include the equation, calculated concentration, and one limitation.

Answer key

Section 1. Variables: (a) independent variable = copper sulfate concentration; dependent variable = absorbance. (b) independent variable = nitrate concentration; dependent variable = electrode potential or probe signal. (c) independent variable = glucose concentration; dependent variable = meter signal or displayed glucose reading.

Section 2. For $A = 0.00600C + 0.0113$: (a) $A = 0.00600(75) + 0.0113 = 0.4613$, so about 0.461. (b) blank estimate = 0.0113. (c) $C = (0.493 - 0.0113)/0.00600 = 80.3 \text{ mg L}^{-1}$.

Section 3. Two-point line: slope $m = (0.486 - 0.128)/(80 - 20) = 0.00597$ absorbance units per mg L^{-1} . Intercept $b = 0.128 - (0.00597)(20) = 0.00867$. For $A = 0.300$, $C = (0.300 - 0.00867)/0.00597 = 48.8 \text{ mg L}^{-1}$.

Section 4. Predicted absorbance for 40 mg L^{-1} : $\hat{A} = 0.00600(40) + 0.0113 = 0.2513$. Residual = $0.247 - 0.2513 = -0.0043$. The point is below the line.

Section 5. $R^2 = 0.997$: (a) justified; (b) not justified; (c) not justified; (d) justified.

Section 6. Interpolation/extrapolation: (a) interpolation; (b) questionable or below-range extrapolation, because it is below the blank/lowest standard signal; (c) above-range extrapolation; (d) interpolation.

Section 7. Spreadsheet formulas: (a) =SLOPE(B2:B7,A2:A7). (b) =INTERCEPT(B2:B7,A2:A7). (c) =(B10-\$E\$3)/\$E\$2.

Mini-practical. Regression gives approximately

$$A = 0.00750C + 0.00300, \quad R^2 = 0.99994.$$

For the diluted unknown, $C = (0.268 - 0.00300)/0.00750 = 35.3 \text{ mg L}^{-1}$. The dilution factor is $50.00/10.00 = 5.00$, so the original concentration is $35.3 \times 5.00 = 176.7 \text{ mg L}^{-1}$, reported as about 177 mg L^{-1} . The check sample gives $C = (0.421 - 0.00300)/0.00750 = 55.7 \text{ mg L}^{-1}$, which is outside the 0 mg L^{-1} to 50 mg L^{-1} standard range; it should be diluted or the calibration range should be extended. Possible improvements: prepare more standards near the unknown, include replicate standards, check residuals, use a fresh blank, or dilute samples that exceed the highest standard.

Reporting conclusion. A suitable answer should state that the unknown from Section 3 has concentration about 55.1 mg L^{-1} using $A = 0.00600C + 0.0113$, and should mention a limitation such as reliance on the linear range, blank accuracy, or standard preparation quality.